



HACKTHEBOX

Neurogrid CTF: Human-Only

Reversing :: IronHeartEcho

1. Quick view of the downloaded file:
: `ls -l ./rev_ironheart_echo.zip`
2. We decompress in silent mode:
: `unzip -q ./rev_ironheart_echo.zip`
3. We display the structure and enter the folder:
: `tree && cd ./rev_ironheart_echo`

```
(uglyB4sTRD@cheapPC)-[~/Desktop]
$ ls -l ./rev_ironheart_echo.zip
-rw-r--r-- 1 uglyB4sTRD uglyB4sTRD 3499 Nov 27 00:24 ./rev_ironheart_echo.zip

(uglyB4sTRD@cheapPC)-[~/Desktop]
$ unzip -q ./rev_ironheart_echo.zip

(uglyB4sTRD@cheapPC)-[~/Desktop]
$ tree && cd ./rev_ironheart_echo
.
├── rev_ironheart_echo
│   └── iron
└── rev_ironheart_echo.zip

2 directories, 2 files

(uglyB4sTRD@cheapPC)-[~/Desktop/rev_ironheart_echo]
$
```

4. Run and test the binary:

```
: ./iron
```

5. **ltrace** 'It is useful for debugging/understanding a binary without having its source code':

```
: ltrace ./iron
```

6. Our input **TeSt_string_HERE** is compared with **strcmp()**, we find the flag in the second argument...!!!

```
"HTB{r3wr1tt3n_r3s0nanc3}"
```

```
(uglyB4sTRD@cheapPC)-[~/Desktop/rev_ironheart_echo]
$ ./iron
[Core] Resonance mismatch.
[Sentinel] Purpose altered. Cannot authenticate echo.
Enter resonance pattern:
> TeSt_string_HERE
Pattern rejected. Echo lost.

(uglyB4sTRD@cheapPC)-[~/Desktop/rev_ironheart_echo]
$ ltrace ./iron
puts("[Core] Resonance mismatch.") = 27
puts("[Sentinel] Purpose altered. Cann... [Sentinel] Purpose altered. Cannot authenticate echo.
) = 54
printf("Enter resonance pattern:\n> "Enter resonance pattern:
) = 27
fgets(> TeSt_string_HERE
"TeSt_string_HERE\n", 128, 0x7f9337f268e0) = 0x7ffd3c428360
strlen("TeSt_string_HERE\n") = 17
memset(0x7ffd3c4282c0, '\0', 25) = 0x7ffd3c4282c0
strcmp("TeSt_string_HERE", "HTB{r3wr1tt3n_r3s0nanc3}") = 12
puts("Pattern rejected. Echo lost."Pattern rejected. Echo lost.
) = 29
+++ exited (status 0) +++

(uglyB4sTRD@cheapPC)-[~/Desktop/rev_ironheart_echo]
$
```